RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Operations Research (Contract), STMicroelectronics (NUS System Design Project) **Aug 2026 – Present**

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories **Jun 2026 – Present**

- Design a tool loop where a model emits schema-constrained JSON and the solver computes every number, gated by an automated numeric-fidelity check and a per-call audit log with no write path to plant control.

Robotics Vision Engineer (AI+X Global Challenge), Amazon **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

Data Analytics & Product Management Intern, Abbott Laboratories **Jan 2026 – Jun 2026**

- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Productionized a team-built APC simulator for live internal users on Azure App Service with Docker packaging, access restrictions, and runtime and session hardening.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defense Tech Hackathon (1,300+ applicants) **Sep 2026**

- Implemented a deterministic Unreal Engine 5.8 C++ simulation with Claude Code; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, against 2.1 for an operator-run fleet.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.

Competitor, DSTA BrainHack 2025 CTF **Feb 2025**

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, finishing around 90th of 400+ teams.

LEADERSHIP AND ACTIVITIES

YesWeHack, Independent Bug Bounty Researcher **May 2024 – Present**

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports with Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.
- Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.

NTUC Health Active Ageing Centre, Elderly Care Volunteer **Sep 2025 – Sep 2026**

- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.

Republic of Singapore Navy, Base Support Assistant **Jan 2022 – Jan 2023**

- Compressed a 6-month food-waste analysis cycle into a 3-minute reporting process with an Excel VBA automation workflow inside a security-constrained environment.

SKILLS AND CERTIFICATIONS

Technical: Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, fuzzing, reverse engineering, custom exploit scripting, Python, Bash, Rust, Claude Code, ChatGPT, Gemini, ComfyUI
Certifications: Reverse Engineering Windows Executables (digital forensics); Docker for the Absolute Beginner (Stone River eLearning)